Math Inquirers Primary 5: Scheme of Work

Holiday program for ages 10 to 12 (builds on Primary 5 maths) • 10 lessons of 90 minutes • 2 lessons a week for 5 weeks • live online, up to 8 children

Approach. Enrichment, not extra drill: non-routine puzzles on topics chosen to complement the school syllabus. Lessons 1 to 9 follow the same routine, in minutes of the 90: Opening (2 to 3.5), sneak-peek puzzles (4 to 4.5), warm-up (5.5 to 6.5), explore (12.5 to 14), mini-lesson (11 to 12.5), brain break (4 to 4.5), worksheet page 1 (pairs) (12.5 to 15.5), check page 1 (3.5 to 4), short teaching before page 2 (1.5 to 3.5), worksheet page 2 (alone) (7 to 11.5), check page 2 (2 to 3), talk time (5.5), exit ticket, reflection and close (7 to 9.5), and buffer (floating time) (1 to 5). Lesson 10 has its own plan. The time plan under each lesson is the one in the tutor guide.
Habits. Notice it. Test it. Explain why.

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
1 Week 1A	**Million Puzzles** Numbers to 10 Million	Read, write and compare numbers up to 10 million; solve digit-card and digit-riddle puzzles with place value (testing the best number just below and just above a target); estimate to check large calculations; and use the order of operations and brackets to hit target numbers.	Place value to 10 million, testing both sides of a target, reasoning from clues in order, estimating by rounding, order of operations with brackets, making a list in order	(a) What is the value of the digit 4 in 8 406 125? (b) Add one pair of brackets to make it true: 5 + 3 × 8 − 2 = 62
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6** · Explore **14** · Mini-lesson **11** · Brain break **4.5** · Worksheet page 1 (pairs) **14.5** · Check page 1 **4** · Short teaching before page 2 **3** · Worksheet page 2 (alone) **8** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **8** · Buffer (floating time) **2**			
2 Week 1B	**Fraction Feats** Fractions: Four Operations	Link fractions and division (3 ÷ 4 = 3/4) and (a puzzle that goes further) divide a proper fraction by a whole number; write fractions as the sum of two different unit fractions by testing in order; solve fraction-of-a-remainder problems forwards and backwards with a bar model; and spot and explain patterns in fraction sums.	Fraction and division, drawing a model (bar, pizza), testing in order and knowing when to stop, fraction of a remainder, working backwards, looking for a pattern, testing both ways	(a) 4 pizzas are shared equally among 5 children. How much does each child get? (b) Ali had 60 marbles. He gave 1/3 of them to Mei and 1/2 of the remainder to Ben. How many marbles did Ali have left?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6.5** · Explore **14** · Mini-lesson **11** · Brain break **4** · Worksheet page 1 (pairs) **15.5** · Check page 1 **4** · Short teaching before page 2 **1.5** · Worksheet page 2 (alone) **8** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **8** · Buffer (floating time) **2.5**			
3 Week 2A	**Before and After** Fraction Word Problems	Solve fraction word problems with units and before-and-after models: fraction of a remainder, giving between two people (the total stays the same), both getting or spending the same amount (the difference stays the same), and one amount that does not change.	Drawing bar models, choosing what stays the same, splitting parts into equal units, working backwards, checking an answer against every condition	Ali had 35 stickers and Siti had 25. How many stickers must Ali give Siti so that Ali has 1/2 as many stickers as Siti?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6.5** · Explore **13.5** · Mini-lesson **12** · Brain break **4** · Worksheet page 1 (pairs) **14** · Check page 1 **3.5** · Short teaching before page 2 **1.5** · Worksheet page 2 (alone) **8** · Check page 2 **2** · Talk time **5.5** · Exit ticket, reflection and close **9.5** · Buffer (floating time) **3.5**			
4 Week 2B	**Triangle Areas** Area of Triangle	Find the area of a triangle as 1/2 × base × height, choosing a base and the height that meets it at a right angle (also when the height is outside the triangle); use “same base, same height” to explain equal areas; find areas of composite figures and shaded parts by splitting into triangles or by taking away.	Spotting the height, ignoring lengths that are not needed, reasoning with same base and same height, splitting a figure, taking away from a rectangle, explaining why an area does not change	A triangle has an area of 42 cm². Its base is 12 cm. What is its height?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6** · Explore **13** · Mini-lesson **11** · Brain break **4.5** · Worksheet page 1 (pairs) **12.5** · Check page 1 **3.5** · Short teaching before page 2 **3** · Worksheet page 2 (alone) **10** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **7** · Buffer (floating time) **4.5**			

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
5 Week 3A	**Volume Vault** Volume	Find the volume of a cuboid by counting layers of 1-cm cubes and by length × breadth × height; list every cuboid that can be built from a number of cubes; solve water-tank problems (cm³ and litres, new water levels, the rise when an object sinks, pouring from one tank into another); and count the painted faces of a cube cut into small cubes.	Make a list in order, draw a diagram to scale, work backwards (height = volume ÷ base area), look for a pattern (painted cubes), check with a second method	A tank has a base 30 cm by 25 cm and water 8 cm deep. A rock is put in and sinks. The water rises to 10 cm. (a) What is the volume of the rock? (b) How many litres of water are in the tank?
	Time plan (minutes): Opening **3.5** · Sneak-peek puzzles **4.5** · Warm-up **6.5** · Explore **13.5** · Mini-lesson **12** · Brain break **4.5** · Worksheet page 1 (pairs) **15** · Check page 1 **3.5** · Worksheet page 2 (alone) **7** · Check page 2 **2** · Talk time **5.5** · Exit ticket, reflection and close **8.5** · Buffer (floating time) **4**			
6 Week 3B	**Rate Racers** Rate	Find a rate for 1 unit and use it to scale up or work backwards in a rate table; compare buys by the price of one item; add rates when taps or printers work together and take away a leak; work out charges “for every half hour or part of it”; and compare two price plans with a table.	Find the value of 1 unit, make a table, compare, guess and check with a table, draw a timeline, check with a second method	Tap A fills 15 ℓ a minute and tap B fills 10 ℓ a minute. How long do both taps together take to fill a 200 ℓ tank? How long if a leak also lets out 5 ℓ a minute?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4** · Warm-up **6** · Explore **14** · Mini-lesson **12.5** · Brain break **4** · Worksheet page 1 (pairs) **14.5** · Check page 1 **4** · Short teaching before page 2 **1.5** · Worksheet page 2 (alone) **9** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **8.5** · Buffer (floating time) **1.5**			
7 Week 4A	**Percentage Puzzles** Percentage	Find any percentage of an amount by building it from 10%, 5% and 1%; work out a discount and then 9% GST on the sale price; find a percentage of a remainder; and spot the classic percentage traps. (A puzzle that goes further, on the Think harder page: find a percentage increase or decrease by comparing the change with the original amount.)	Asking "percentage of what?", building from 10% and 1%, bar models, making a table, comparing deals, explaining a trap	A $30 toy is sold at 20% off. Then 9% GST is added to the sale price. How much do you pay?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **5.5** · Explore **12.5** · Mini-lesson **11** · Brain break **4.5** · Worksheet page 1 (pairs) **14** · Check page 1 **3.5** · Short teaching before page 2 **1.5** · Worksheet page 2 (alone) **11.5** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **9** · Buffer (floating time) **2**			
8 Week 4B	**Angle Chasers** Angles, Triangles and Quadrilaterals	Use the angle facts (straight line 180°, point 360°, vertically opposite, triangle 180°, isosceles and equilateral triangles, parallelogram, rhombus, trapezium) to chase unknown angles through a figure, step by step, giving a reason for every step; and list every case when only one angle of an isosceles triangle is given.	Angle-chase chains with reasons, using a diagram carefully, finding all cases, checking by adding, explaining	In triangle PQR, PQ = PR and angle QPR = 44°. Find angle PQR. Write the reason for each step.
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4** · Warm-up **6** · Explore **12.5** · Mini-lesson **12** · Brain break **4.5** · Worksheet page 1 (pairs) **13** · Check page 1 **3.5** · Short teaching before page 2 **3** · Worksheet page 2 (alone) **7.5** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **8.5** · Buffer (floating time) **5**			
9 Week 5A	**Pattern Generalisers** Patterns and Number Sense	Continue growing patterns drawn as figures, say the rule for any figure in words (for example, number of sticks = 3 × figure number + 1) and explain it from the picture, work backwards from a total to find the figure, and solve “two kinds of things” problems with a guess-and-check table and with supposition.	Looking for a pattern, making a table, describing the rule in words, working backwards, guess and check, supposition, explaining from a picture	Figure 1 of a pattern has 6 dots. Each figure has 4 more dots than the figure before. (a) How many dots are in figure 20? (b) Which figure has 102 dots?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **5.5** · Explore **13.5** · Mini-lesson **11.5** · Brain break **4** · Worksheet page 1 (pairs) **14** · Check page 1 **4** · Short teaching before page 2 **3.5** · Worksheet page 2 (alone) **9** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **9** · Buffer (floating time) **1**			
10 Week 5B	**Challenge Day** Mixed Strategies	Choose a way to start a puzzle (draw a model, list in order, make a table, work backwards, guess and check or suppose, use a logic grid), solve competition-style problems, check every clue, and explain why the chosen start works.	Choosing a strategy, working backwards with fractions, logic grids, models with units, systematic listing, supposition, checking against every clue, explaining	Ravi spends 1/2 of his money on a ticket, then $5 on a snack, and has $7 left. How much money did he start with? Say which start you used.
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6.5** · Explore **14.5** · Mini-lesson **10.5** · Brain break **4** · Worksheet page 1 (pairs) **13.5** · Check page 1 **3** · Short teaching before page 2 **2** · Worksheet page 2 (alone) **7.5** · Check page 2 **3** · Talk time **5.5** · Exit ticket, reflection and close **10.5** · Buffer (floating time) **3**			

Assessment and progress

Each lesson has a 2-page Start here worksheet, then Think harder, Go further and an Investigation with a Look-back (printed for every child, who may choose any of them at any time; nobody is placed in a group), an exit ticket and an Inquiry Journal stamp. Tutors record S (secure), D (developing) or H (needs help) for each child on the progress tracker in the tutor guide. Parents receive a short note each week and the Weekly Dose of Thinking Math practice pack.

Materials

Printed student workbook with cut-outs, pencil, scissors and paper; a laptop or tablet with a camera. Tutors use the slides for each lesson and the tutor guide scripts.